# Constructing and Validating an Agent-Based Simulator for Uniswap v3 Concentrated Liquidity

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**Abstract**

We construct a quantitatively interpretable agent-based simulator of a Uniswap v3 concentrated-liquidity pool and validate it against six months of Ethereum mainnet data for the WETH/USDC 5 bps pool. The simulator represents liquidity providers by finite tick ranges, executes swaps with the square-root price update of concentrated liquidity, removes positions through burns, and allocates fees only to active liquidity. Every calibrated parameter admits a market interpretation: the initial price fixes the pool state; tick spacing defines the admissible range grid; the fee tier governs LP income; the training-derived range widths and sizes, together with a nested range ladder and a periodically re-centred core sleeve, control capital concentration; and a conditional Student-$t$ jump diffusion governs aggregate order flow.

The empirical sample comprises 181 daily OHLCV bars from 2026-01-22T00:00:00 UTC to 2026-07-21T00:00:00 UTC, with recorded dollar volume of approximately $24.79 billion, together with the complete indexed mint/swap/burn stream. Calibration is chronological: price-process and structural LP parameters are fit on the first 120 days and frozen for a 61-day holdout evaluated over 6 independent paths. The calibrated model uses 8 compressed event slots per daily bar, multi-LP mint bundles (mean size 12), empirically informed widths, a core sleeve re-centred every 8 slots, state-dependent burns, and LP refreshes. The diffusion scale is daily and is converted to slot variance by the square-root time rule; slot and daily support caps prevent a single compressed event from creating an implausible price jump. Diffusion scale, degrees of freedom, jump probability and jump scale are $s = 0.028$, $\nu = 3.5$, $\lambda = 0.500\%$ and $\sigma_J = 0.08$.

Out of sample, the pooled daily log-return standard deviation is 0.0301 against 0.0292 empirically, while the simulated and empirical mean high-low ranges are 0.0291 and 0.0411, respectively, and excess kurtosis is 3.6994 versus 3.4888. Protocol diagnostics remain at floating-point precision for the concentrated-liquidity invariant. Distributional distances on a representative path are KS = 0.1000 and Wasserstein = 0.0058. We report an explicit bootstrap ablation and a fixed-time liquidity-by-tick comparison (correlation 0.8568). The simulator is therefore a mechanistic baseline rather than a price-forecasting engine: every state transition preserves the Uniswap v3 equations, while remaining volatility and inactive-liquidity gaps are stated explicitly.

**Keywords:** Uniswap v3; concentrated liquidity; automated market makers; agent-based simulation; market microstructure; Monte Carlo validation.

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# 1 Introduction

Automated market makers (AMMs) replace a limit-order book with a deterministic trading curve. In constant-function market makers, liquidity supplied by users defines the curve against which trades are executed [3, 4]. Uniswap v2 implements a global constant-product invariant [1], whereas Uniswap v3 applies the same economic idea locally over finite price intervals [2]. The central innovation of concentrated liquidity is that a liquidity provider (LP) chooses a range $[p_a, p_b]$ and earns fees only while the pool price remains inside that range [9, 13, 19].

The purpose of this paper is deliberately narrow. We do not forecast the WETH/USDC price path. Instead we build a market *mechanism* that (i) follows the Uniswap v3 equations at the position level, (ii) produces realistic pool-level return statistics, and (iii) exposes every state variable required for statistical and protocol validation. This objective matches the validation philosophy of agent-based financial-market simulation: a model should reproduce robust stylised facts, not a single historical trajectory pointwise [6, 8, 11, 15].

Validation therefore has two layers.

1. **Mechanical.** After every event the simulator reconstructs real reserves, active and inactive liquidity, accumulated fees, execution slippage, and the relative residual of the Uniswap v3 invariant.
2. **Statistical.** A six-month daily dataset from the WETH/USDC 5 bps pool is split chronologically into calibration and holdout periods to test whether simulated returns reproduce volatility, high-low range, heavy tails and distributional distances.

A simulator can match returns while violating the protocol, or satisfy the invariant while generating unrealistic markets. A useful baseline must pass both checks.

**Contribution.** Relative to prior AMM analytics and DeFi characterisation studies, we contribute: (i) an event-driven simulator whose mint/swap/burn mixture is estimated from the on-chain stream rather than imposed arbitrarily; (ii) a two-stage chronological calibration that separates price shocks from LP structure; (iii) protocol diagnostics (invariant residual, active/inactive shares, slippage) reported alongside return statistics; and (iv) a reproducible artefact pipeline that maps every published figure and table to open source scripts (Appendix A).

# 2 Related literature

Constant-function market makers are now a mature theoretical object [3, 4]. Uniswap v2 and v3 white papers specify the invariants and fee accounting used in production [1, 2]. Strategic LP behaviour under concentrated liquidity has been analysed game-theoretically [9], while empirical work documents range placement, fee income and liquidity concentration on Uniswap v3 [13, 19].

Separately, the empirical finance literature emphasises stylised facts of returns, heavy tails, weak linear autocorrelation, volatility clustering, as targets for simulation models [5–7, 12]. Agent-based validation criteria stress distributional and dependence reproduction rather than

path matching [8, 11, 15]. Range-based volatility estimators motivate our use of the high-low statistic alongside close-to-close returns [14].

Our contribution sits at the intersection: we import the stylised-fact validation programme into a protocol-faithful Uniswap v3 simulator, using official fee-tier and concentrated-liquidity conventions [16, 17] and subgraph-indexed events [18].

# 3 Uniswap v3 mechanics

Let $x$ denote the WETH-like reserve and $y$ the USDC-like reserve. A Uniswap v3 position is defined by a lower price $p_a$, an upper price $p_b$ and a liquidity amount $L$. When the current price $P$ lies inside the range, the position satisfies

$$\left(x + \frac{L}{\sqrt{p_b}}\right)(y + L\sqrt{p_a}) = L^2. \tag{1}$$

The added terms are virtual reserves. They make a finite-range position behave locally like a constant-product AMM:

$$x^{\text{virt}} = x + \frac{L}{\sqrt{p_b}}, \qquad y^{\text{virt}} = y + L\sqrt{p_a}, \qquad P = \frac{y^{\text{virt}}}{x^{\text{virt}}}.$$

If the price reaches the upper boundary, the real $x$ reserve is depleted and the position is fully converted into $y$:

$$y^{\text{rem}} = L(\sqrt{p_b} - \sqrt{p_a}).$$

If the price reaches the lower boundary, the position is fully converted into $x$:

$$x^{\text{rem}} = L\left(\frac{1}{\sqrt{p_a}} - \frac{1}{\sqrt{p_b}}\right). \tag{2}$$

This boundary behaviour is the source of the distinction between total liquidity and active liquidity: deposited capital can remain in the pool while being irrelevant for the next trade.

## 3.1 Ticks and decimal scaling

Uniswap v3 discretises admissible range endpoints through ticks. The simulator uses the decimal-adjusted convention

$$P(\text{tick}) = \frac{10^{12}}{1.0001^{\text{tick}}}. \tag{3}$$

The factor $10^{12}$ corrects the decimal difference between WETH (18 decimals) and USDC (6 decimals). The base 1.0001 is the standard tick granularity: before decimal adjustment, one tick is a multiplicative price change of roughly one basis point. The simulated pool uses the 5 bps fee tier, $f = 0.0005$, and tick spacing 10, matching a highly liquid WETH/stablecoin pool [16, 17].

| Phase | Events | Mint | Swap | Burn |
|---|---|---|---|---|
| bootstrap | 78208 | 1.8911% | 95.9250% | 2.1839% |
| operating | 703878 | 2.7530% | 94.2260% | 3.0210% |

Table 1: Event probabilities estimated without holdout leakage. The bootstrap contains the first 10% of the chronological training event stream; the operating regime contains the remaining 90%.

# 4 Simulator construction

At time $t$, the state is

$$S_t = \{P_t, \text{tick}_t, \mathcal{P}_t, F_t\},$$

where $P_t$ is the pool price, $\text{tick}_t$ is the current tick, $\mathcal{P}_t$ is the set of open liquidity positions, and $F_t$ is cumulative fee income. Each position $j$ is represented by $(\text{tick}_{\ell,j}, \text{tick}_{u,j}, L_j)$. These three quantities are sufficient to reconstruct the position real reserves at any price. For an active position,

$$x_j(P) = L_j \left( \frac{1}{\sqrt{P}} - \frac{1}{\sqrt{p_b}} \right), \qquad y_j(P) = L_j \left( \sqrt{P} - \sqrt{p_a} \right). \tag{4}$$

Below the range the position is entirely in $x$; above the range it is entirely in $y$. This reconstruction is performed after every event and is also used to compute the invariant residual in Eq. (1).

## 4.1 Event process

The simulator is a discrete-event model with three requested event-slot types: mint, swap and burn. The first 10% of training events form a bootstrap phase with probabilities

$$p_{\text{mint,swap,burn}} = (1.8911\%, 95.9250\%, 2.1839\%),$$

equal to the observed frequencies in the first 10% of the chronologically ordered training event stream. The first synthetic event is forced to be a mint so that the initial state is non-empty. After bootstrap, the operating regime is

$$p_{\text{mint,swap,burn}} = (2.7530\%, 94.2260\%, 3.0210\%),$$

estimated from the remaining 90% of the training stream (Table 1). In the validation, 8 such compressed event slots are aggregated into one synthetic daily bar. This is a fixed computational compression of the full stream, which contains thousands of events per day; it is not an assertion that eight slots reproduce the original event count (Appendix A).

### 4.2 Mint

A mint arrival is modelled as a *bundle* of independent LP ranges rather than a single position. The mean bundle size is $\bar{n}_{\mathrm{LP}} = 12$ (Poisson). A fraction $\phi_c = 0.50$ of the legs form a nested staircase of increasing half-widths centred on the current usable tick; the remaining legs are lightly offset shoulders. This construction is what produces the stepwise $L(\text{tick})$ profile in Figure 6: each range endpoint adds a discontinuity, and many overlapping LPs accumulate into a ladder. Stranded inactive ranges whose midpoints lie more than 1200 ticks from the current tick are pruned, and the book is capped at 350 positions, so historical mountains do not dominate the local book. These are explicit reduced-form LP migration rules.

Shoulder centres use partial anchoring

$$c_t = \mathrm{round}\big[t_t + (1-a)\widetilde{c}_t\big], \qquad a = 1, \tag{5}$$

where $\widetilde{c}_t$ is a training-window empirical centre offset clipped to a local window. The training mint-width distribution (median 36.0, 90th percentile 580.0) informs admissible widths, while the nested peak legs use a compact deterministic ladder so that the normalised maximum stays near the snapshot tick.

The nested peak legs are marked as the *core sleeve*. At the beginning of each 8-slot interval, every core range is translated by the difference between the current usable tick and its previous centre. The translation preserves width, liquidity, reserves and accrued fees; the ledger records a negative old range and a positive new range. Consequently the dominant simulated $L(\text{tick})$ profile is centred at offset zero at each interval snapshot, while non-core shoulders and wider ranges are allowed to remain inactive until the burn/refresh rules remove or replace them. This is a mechanistic representation of active market-making capital, not an ex post translation of the plotted curve.

For standalone experiments without an empirical sample, the fallback width law

$$w = m + (M-m)e^{-dr}, \qquad r \sim U(0,1), \tag{6}$$

with $m = 2$, $M = 10000$ and $d = 8$ is available. The exponential decay places higher probability on narrow ranges near the current price [13, 19]. In the calibrated model the training-derived distribution replaces this fallback for shoulder and rebalanced ranges; nested peak legs use the deterministic ladder described above.

Given a deposit of token $x$, minted liquidity is

$$\Delta L = \Delta x \frac{\sqrt{P}\sqrt{p_b}}{\sqrt{p_b} - \sqrt{P}},$$

and the required amount of token $y$ is $\Delta y = \Delta L(\sqrt{P} - \sqrt{p_a})$. The values 1 during bootstrap and 0.1 afterwards are base WETH-like notional scales for a mint bundle. Per-leg staircase weights and training-derived size multipliers determine the realised aggregate deposit, so these values are not interpreted as exact total mint notionals.

### 4.3 Swap

Only active liquidity participates in a swap:

$$L_{\text{act}}(t) = \sum_{j:P_t \in [p_{a,j}, p_{b,j}]} L_j.$$

For a net input $\Delta x_t$, the square-root price update is

$$\sqrt{P_{t+1}} = \frac{1}{1/\sqrt{P_t} + \Delta x_t / L_{\text{act}}(t)}. \tag{7}$$

The output of token $y$ is $\Delta y_t = -L_{\text{act}}(t)\big(\sqrt{P_{t+1}} - \sqrt{P_t}\big)$.

For statistical validation, the aggregate price shock is generated before the swap and then converted into the $\Delta x_t$ required by Eq. (7). Specifically,

$$\begin{aligned} h_t &= \omega + \alpha r_{t-1}^2 + \beta h_{t-1}, && \omega = s^2(1-\alpha-\beta), \\ r_t &= \sqrt{h_t}\, z_t + J_t, && z_t \sim t_\nu, \quad J_t \sim \text{Bernoulli}(\lambda)\,\mathcal{N}(0, \sigma_J^2), \\ P_{t+1} &= P_t e^{r_t}. \end{aligned} \tag{8}$$

The Student-$t$ innovation is standardised to unit variance when $\nu > 2$. We fix $\alpha = 0.06$ and $\beta = 0.90$. Here $s$ is a daily diffusion scale and the slot innovation uses $s_{\text{slot}} = s/\sqrt{m}$, with $m = 8$. Each slot shock is clipped to $[-0.08, 0.08]$ and the cumulative move within a daily bar is clipped to $[-0.18, 0.18]$ relative to the bar open. These are support guards for the compressed time discretisation, not additional jump mechanisms. Calibrated values are $s = 0.028$, $\nu = 3.5$, $\lambda = 0.500\%$ and $\sigma_J = 0.08$. This construction combines GARCH-style volatility clustering with a separate mechanism for discontinuous flow [5–7]. Given the target $P_{t+1}$, implied order flow is

$$\Delta x_t = L_{\text{act}}(t)\left(\frac{1}{\sqrt{P_{t+1}}} - \frac{1}{\sqrt{P_t}}\right).$$

Fees equal $f|\Delta x_t|P_t$ and are allocated pro rata to positions active before the trade.

### 4.4 Burn and refresh

A burn closes one open position, returns its current reserves from Eq. (4), and realises accumulated fees. Position $j$ is selected with weight

$$\begin{aligned} w_j = &\Big[1 + \min\big((t - t_j^{\text{mint}})/\tau_b, 4\big)\Big] \\ &\times \big[1 + (B-1)\mathbf{1}\{P_t \notin [p_{a,j}, p_{b,j}]\}\big] \\ &\times \big[1 + \log(1 + L_j/\,\text{median}(L))\big]. \end{aligned} \tag{9}$$

The calibrated burn-age scale is $\tau_b = 8$ event slots and the inactive-position multiplier is $B = 8$: an inactive position receives exactly $B$ times the activity component of an active position. With probability 100.0% an inactive burn is followed by a re-centred micro-bundle with the same aggregate liquidity. During each swap slot, up to 2 eligible inactive positions independently

enter a refresh trial with probability $q = 60.0\%$ per trial. Refreshes and pruning are recorded as internal burn+mint transitions. They do not alter the calibrated distribution of requested event slots, although they do add state transitions within a slot. Because one synthetic slot compresses many on-chain events, $q$ is a probability per compressed slot and must not be read as the probability attached to one observed Ethereum swap.

## 5 Empirical dataset

The benchmark is the Uniswap v3 WETH/USDC 5 bps pool on Ethereum. Daily OHLCV observations (181 bars from 2026-01-22T00:00:00 UTC to 2026-07-21T00:00:00 UTC) are obtained from the GeckoTerminal API [10]. Total recorded dollar volume is approximately \$24.79 billion. We use close-to-close log returns $r_t = \log(C_t/C_{t-1})$ and the high-low range $h_t = \log(H_t/L_t)$, the latter being closely related to Parkinson estimator [14]. The first 120 days are used for calibration and the subsequent 61 days are held out.

The event benchmark contains the complete indexed stream for the same pool and horizon: 1,083,191 swaps, 27,132 mints and 33,871 burns. GraphQL entity definitions follow the official Uniswap subgraph documentation [18]. OHLCV series calibrate the price process; the event stream estimates event-mixture probabilities and empirical LP range and notional distributions.

## 6 Calibration and statistical validation

Calibration is chronological and has two stages.

1. **Price process.** On the training window we minimise a relative loss over daily volatility, high-low range and excess kurtosis on a grid over $(s, \nu, \lambda, \sigma_J)$. A one-seed screening evaluates the reduced grid; the six lowest-loss cells plus one explicitly tail-heavy candidate are then re-evaluated over four additional training seeds. The final cell is selected by a training-only kurtosis gap that weights the full-window gap by 75% and the second-half training-tail gap by 25%, with refined loss as a tie-breaker. The full-window weight reduces finite-sample sensitivity to a few extreme observations while retaining an explicit tail check. The holdout is never used in this selection.

2. **LP structure.** On the same window we fit $(a, B, \tau_b, p_{\text{reb}}, q)$ to active/inactive liquidity shares and the mean distance of open position centres from the current tick. Because this grid is estimated from one short training path, configurations within 10% of the minimum structural loss are treated as equivalent and the least aggressive burn/age scales are retained.

Event probabilities and empirical mint width, centre and size distributions are estimated using only timestamps before the holdout boundary. The calibrated simulator is then evaluated over 6 independent holdout paths. Each path is simulated over the full window so that liquidity, position ages and fees are carried across the train/test boundary. At the boundary the absolute price is re-anchored to the first observed holdout close and all position ticks are translated by the

| Parameter | Value | Interpretation |
|---|---|---|
| $s$ | 0.028 | diffusion scale |
| $\nu$ | 3.5 | Student-$t$ degrees of freedom |
| $\lambda$ | 0.500% | jump probability per swap slot |
| $\sigma_J$ | 0.08 | jump standard deviation |
| $a$ | 1 | shoulder-range tick anchoring |
| $B$ | 8 | inactive burn-weight multiplier |
| $\tau_b$ | 8 | burn-age scale in event slots |
| $p_{\text{reb}}$ | 100.0% | inactive-burn re-mint probability |
| $q$ | 60.0% | refresh probability per eligible trial |
| $\bar{n}_{\text{LP}}$ | 12 | mean LP legs per mint slot |
| $\phi_c$ | 0.50 | fraction of mint legs in the centred core |
| $m$ | 8 | compressed slots per daily bar |
| $c_s$ | 0.08 | absolute slot-return cap |
| $c_d$ | 0.18 | absolute daily-bar return cap |
| $\Delta_c$ | 8 | core re-centring interval in slots |
| $w_{\text{sh,max}}$ | 300 | empirical shoulder half-width cap |
| $o_{\text{sh,max}}$ | 500 | empirical shoulder offset cap |

Table 2: Parameters selected by the training-only price and LP-structure calibrations. Percentages are probabilities per eligible event slot or trial, as indicated.

| Metric | Real holdout | MC pooled | Mean path stat. 95% CI |
|---|---|---|---|
| logret std | 0.0292 | 0.0301 | [0.0265, 0.0332] |
| realized vol | 0.2261 | 0.2332 | [0.2053, 0.2569] |
| hl range mean | 0.0411 | 0.0291 | [0.0267, 0.0321] |
| logret kurtosis excess | 3.4888 | 3.6994 | [2.0169, 5.5875] |
| q01 | -0.0905 | -0.0794 | [-0.0868, -0.0579] |
| q99 | 0.0679 | 0.0875 | [0.0659, 0.0912] |

Table 3: Out-of-sample Uniswap v3 benchmark and calibrated Monte Carlo validation. The MC column pools daily returns from all paths; the final column is a 95% bootstrap confidence interval for the mean path-level statistic.

same tick difference; widths, liquidity and relative range locations are preserved. The artificial boundary return is excluded from both samples. The comparison is distributional [8, 15].

Table 3 shows that the refined model produces mean high-low range 0.0291 versus 0.0411 real and test excess kurtosis 3.6994 versus 3.4888. The kurtosis gap is therefore small relative to the empirical tail magnitude, while close-to-close volatility is 0.0301 versus 0.0292 and the high-low range remains understated. The jump component is a calibrated stress mechanism rather than a claim of exact price forecasting. Note that path-level kurtosis intervals need not cover the pooled kurtosis: pooling concatenates independent paths and reweights tail events. Table 4 is the leakage-controlled result: structural fit gives inactive liquidity 0.0462 versus 0.4469 in training and 0.0185 versus 0.3864 in the holdout. Cross-venue arbitrage latency, liquidation cascades and strategic LP withdrawals remain outside the current state transition.

Table 5 reports direct distributional distances. The Kolmogorov-Smirnov statistic is 0.1000 ($p = 0.9284$ on the representative path) and the Wasserstein distance is 0.0058. The represen-

| Phase | Real $\sigma_r$ | Sim. $\sigma_r$ | Real excess $K$ | Sim. excess $K$ | Real inactive $L$ | Sim. inactive $L$ |
|---|---|---|---|---|---|---|
| train | 0.0370 | 0.0291 | 3.6559 | 3.7285 | 0.4469 | 0.0462 |
| test | 0.0292 | 0.0301 | 3.4888 | 3.6994 | 0.3864 | 0.0185 |

Table 4: Train-only calibration and frozen-state out-of-sample test. The simulator carries its LP state from the training window into the holdout; the first boundary return is omitted in both series.

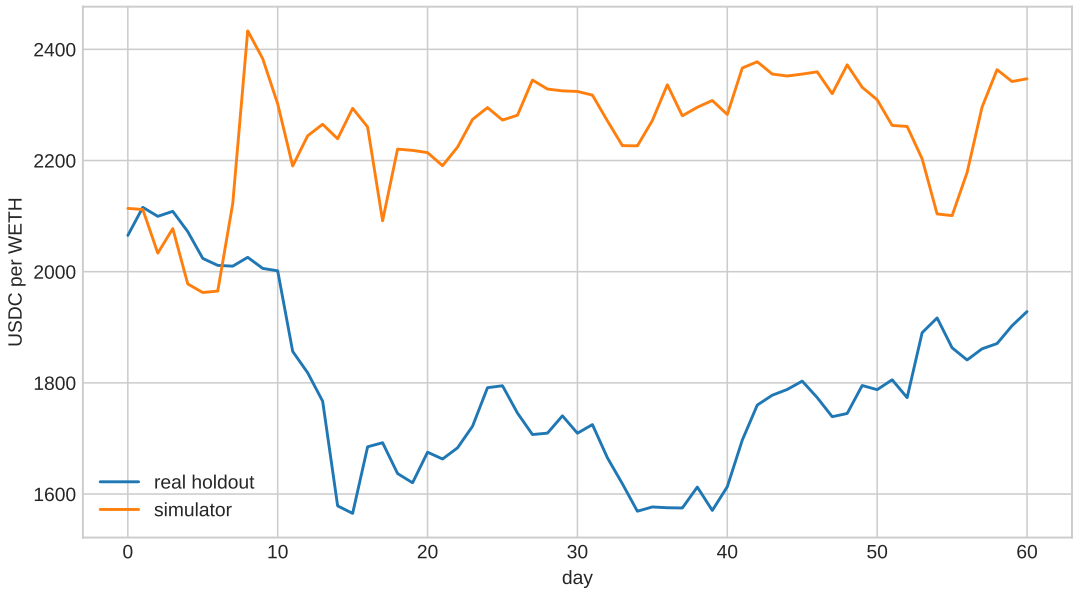



Figure 1: Real holdout close prices and one calibrated simulated path. The goal is distributional similarity, not pointwise path replication. The figure uses the same calibrated path and holdout definition as the statistical tables.

tative path is useful for visual inspection (Figures 1–3), while the pooled ensemble is the more stable estimator of tail moments (Figure 4). The model therefore passes the protocol checks but overstates pooled holdout excess kurtosis (3.6994 versus 3.4888). It is a stress-capable mechanistic model, not a complete explanation of the realised price process.

### 6.1 Protocol diagnostics

Statistical similarity is not sufficient. Table 6 reports diagnostics computed directly from the v3 equations. The invariant residual is of order $10^{-16}$. Active liquidity is well below total liquidity, confirming that finite ranges matter. Unlike a uniform random burn, the calibrated rule preferentially removes and refreshes stale inactive ranges (Figure 5). Slippage is state-dependent (Figure 8): the same order flow moves the price more when active liquidity is thin.

To test the spatial structure of liquidity, we reconstruct the net position book by applying every mint as a positive range-liquidity increment and every burn as a negative increment; refreshes contribute the corresponding negative old range and positive new range. A difference array at the lower and upper tick endpoints then yields $L(\text{tick})$ without expanding every range into individual ticks. We compare the real pool and the simulator at the fixed holdout date

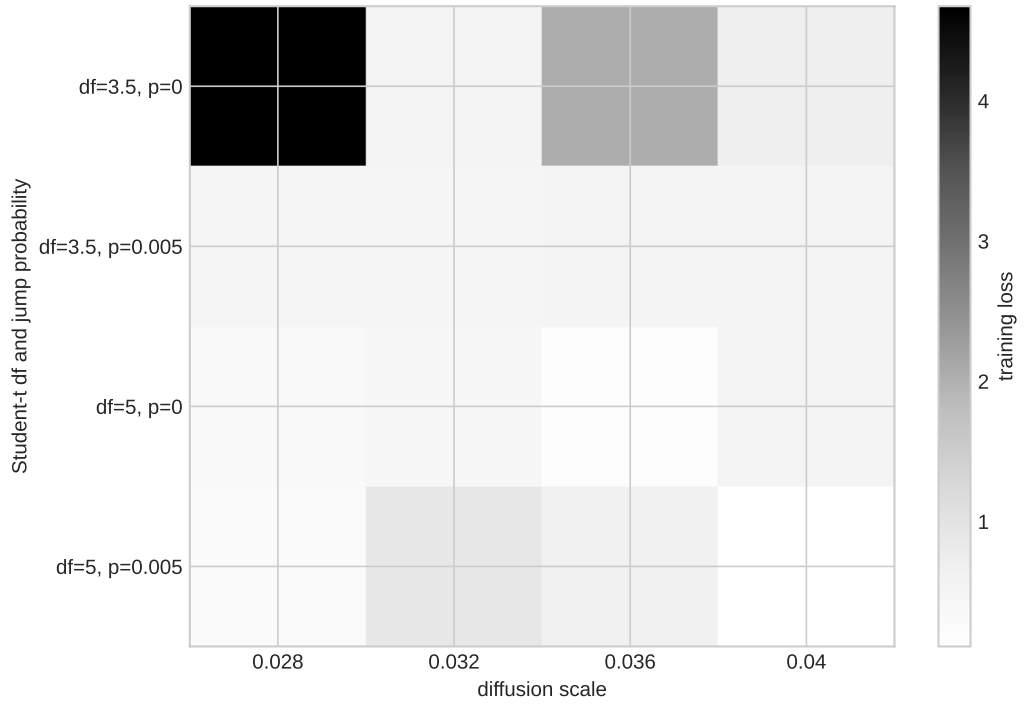



Figure 2: Calibration loss over diffusion scale, Student-$t$ degrees of freedom, jump probability and jump scale. The heatmap takes the best jump scale for each displayed pair in the one-seed screening stage; final selection uses the four-seed refinement of the six best cells plus the explicit tail candidate.

| Statistic | Value |
|---|---|
| ks stat | 0.1000 |
| ks p | 0.9284 |
| wasserstein | 0.0058 |
| volatility ratio | 1.0887 |

Table 5: Distributional fit between empirical daily holdout returns and one calibrated simulated path.

shown in Table 7; the simulated curve is from the representative Monte Carlo seed used in the path diagnostics. Each curve is normalised by its local maximum and expressed relative to its own snapshot tick. The core re-centering rule places the simulated peak at 0 ticks from the current tick, versus -70 in the pool; the normalised profile correlation is 0.8568 and RMSE is 0.2455 (Figure 6). The zero simulated offset is a direct state-level property of the model, whereas the remaining correlation/RMSE gap quantifies the difference in width and shoulder mass rather than a misplaced central peak.

### 6.2 Bootstrap ablation and dependence diagnostics

The initialisation rule is stress-tested separately from the empirical event mixture. Each candidate is evaluated over 5 independent 30-day runs from an empty pool (Table 8, Figure 9). Omitting an explicit depth-building seed produces markedly less active liquidity at the end of

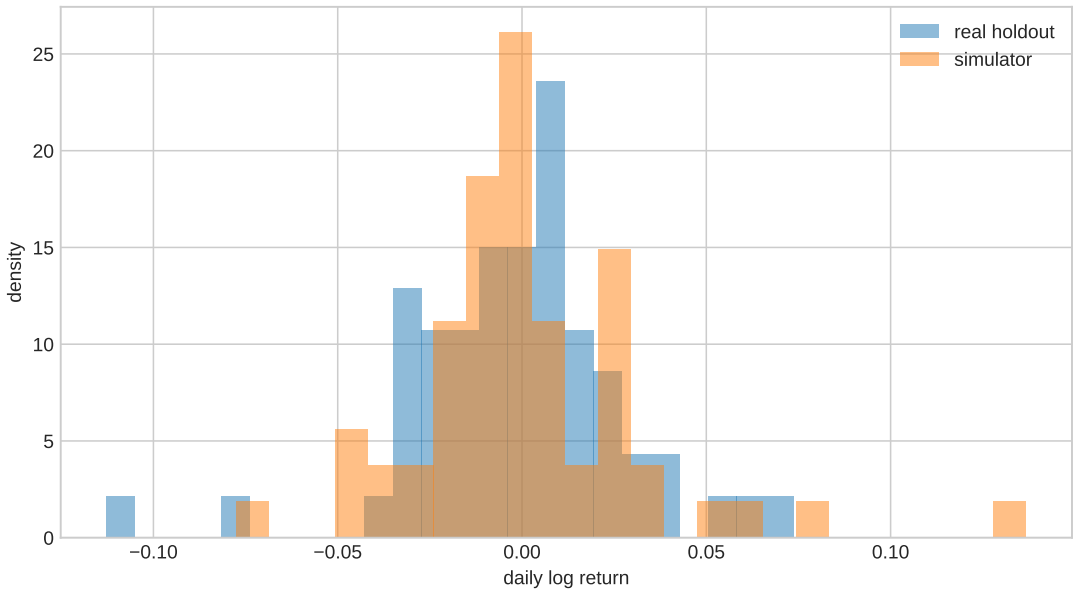



Figure 3: Empirical and simulated daily log-return distributions on the holdout period.

| Diagnostic | MC mean | MC 95% CI |
|---|---|---|
| Active-liquidity share | 0.9815 | [0.9732, 0.9888] |
| Mean absolute slippage | 0.0036 | [0.0032, 0.0040] |
| Mean invariant relative error | 8.623e-17 | [8.381e-17, 8.881e-17] |
| Maximum invariant relative error | 5.340e-16 | [5.042e-16, 5.589e-16] |
| Forced-event share | 0.0000 | [0.0000, 0.0000] |

Table 6: Protocol-level diagnostics from the calibrated simulator.

bootstrap. Forced-event shares remain below 0.2% in every configuration, so the main quantitative effect is initial market depth rather than event feasibility. These candidates do not overwrite the on-chain probabilities in Section 4.1: the production simulator uses the observed mixture and forces only the first synthetic event to be a mint.

Table 9 applies classical stylised-fact tests to daily holdout returns [6, 12]. The real-series Jarque-Bera p-value is 7.564e-07; on the representative simulated path it is 8.883e-16. These path-level tests and the pooled kurtosis answer different questions: a single finite path need not exhibit the same tail evidence as the empirical sample or the pooled ensemble. The Ljung-Box p-values in Table 9 determine whether either series rejects zero linear or absolute-return autocorrelation at lag 10; conclusions are based on the generated table rather than fixed prose.

# 7 Discussion

Three limitations deserve emphasis for a quantitative-finance audience.

**Volatility bias.** The calibrated ensemble holdout volatility bias is 3.1% relative to the real close-to-close standard deviation. The simultaneous high-low comparison is reported in Table 3;

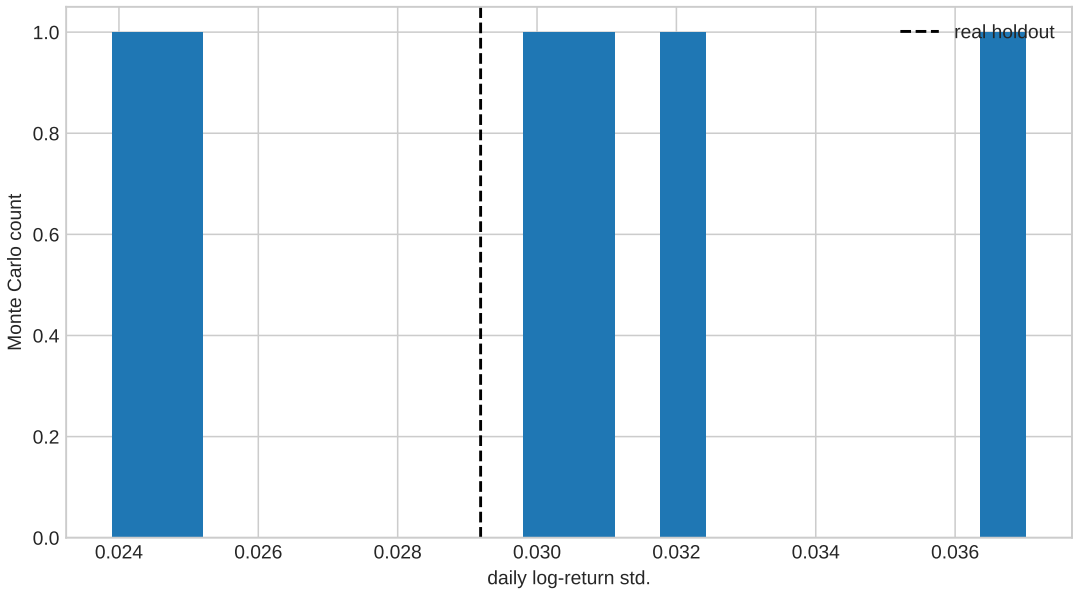



Figure 4: Monte Carlo distribution of simulated daily volatility, with the real pool value shown by the vertical line.

| Metric | Value |
|---|---|
| fixed date utc | 2026-06-21 |
| center tick | 201882 |
| sim center tick | 198817 |
| real peak offset | -70 |
| sim peak offset | 0 |
| normalized rmse | 0.2455 |
| normalized correlation | 0.8568 |
| threshold overlap | 0.5994 |
| sim nonzero steps | 217 |
| real nonzero steps | 668 |

Table 7: Fixed-time liquidity-by-tick comparison. Correlation and RMSE use normalised profiles in a ±5000-tick local window.

whether the two discrepancies have the same sign is therefore determined by the current run, not assumed in advance.

**LP strategy and external markets.** Burns and refreshes are reduced-form. They do not model gas-sensitive withdrawal, inventory hedging, or cross-venue arbitrage that links the pool to CEX mid prices. The inactive-liquidity holdout gap (Table 4) is the quantitative footprint of this omission.

# 8 Conclusion

We construct a Uniswap v3 simulator that generates plausible pool dynamics while preserving concentrated-liquidity equations at the position level. Parameters are economically inter-

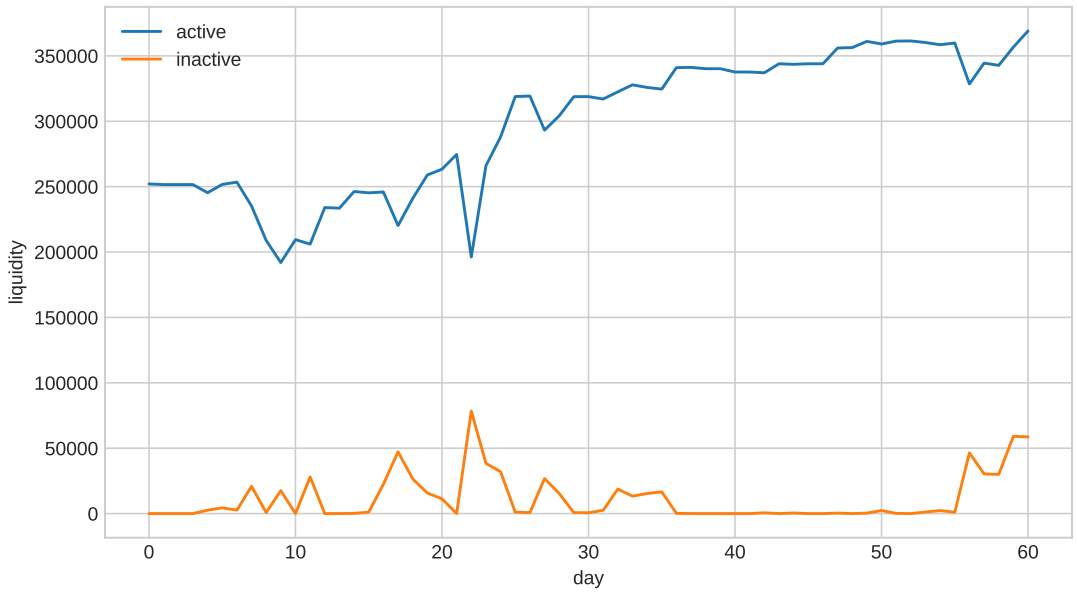



Figure 5: Active and inactive liquidity in one calibrated holdout path.

| Configuration | $L_{act}$ at end | Positions | Forced post events | Executable swaps |
|---|---|---|---|---|
| no bootstrap | 15400.3977 | 13.0000 | 0.0000 | 1.0000 |
| liquidity 0.35 | 445668.4682 | 102.0000 | 0.0000 | 1.0000 |
| selected 0.50 | 601021.2948 | 139.2000 | 0.0000 | 1.0000 |
| liquidity 0.65 | 1094514.1735 | 191.4000 | 0.0000 | 1.0000 |
| long 0.50 | 1495630.6074 | 278.4000 | 0.0000 | 1.0000 |

Table 8: Bootstrap ablation over 5 independent 30-day simulations per configuration. "Forced" means that an infeasible requested action is replaced by a mint to preserve a valid pool state.

pretable; active and inactive liquidity are separated; fees and slippage are endogenous; and the invariant is validated after every event. On a six-month WETH/USDC 5 bps benchmark with chronological holdout validation, the simulator produces a mean high-low range of 0.0291 against 0.0411 empirically, while close-to-close volatility is 0.0301 against 0.0292. The pooled excess kurtosis is close to the empirical value (3.6994 versus 3.4888). State-dependent burns and refreshes reduce inactive-liquidity persistence without eliminating the holdout gap. The fixed-time liquidity profile is centred at zero in the simulated current-price tick and has correlation 0.8568 with residual shape error stated explicitly. Future work should incorporate cross-venue arbitrage, liquidation flow, gas-sensitive withdrawal and richer strategic LP rebalancing. The present model is a mechanistic baseline for quantitative DeFi research.

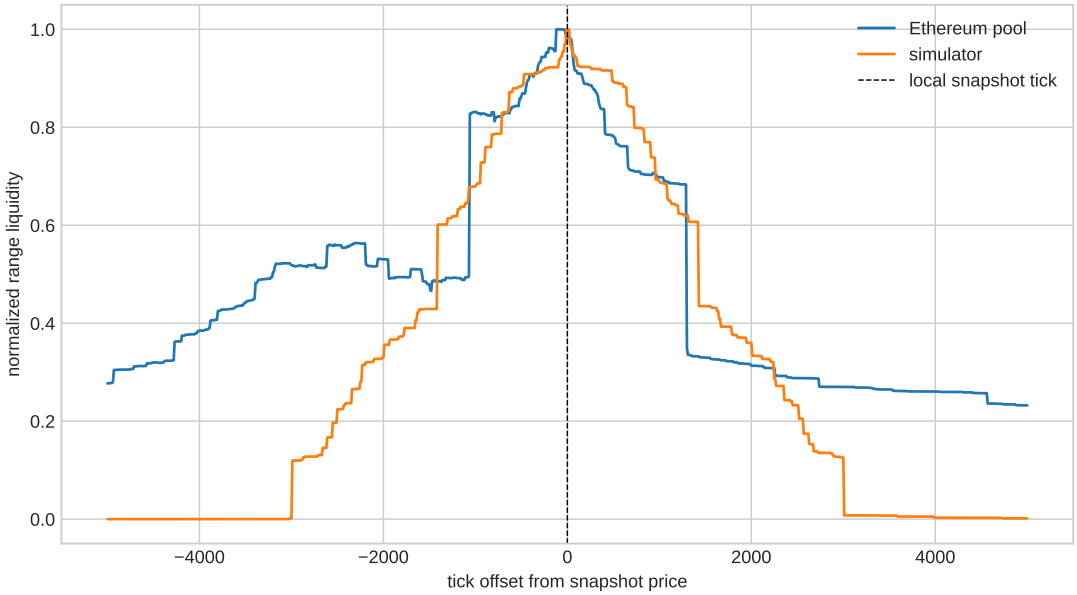



Figure 6: Normalised range liquidity by tick offset at the fixed snapshot. Multi-LP nested mint bundles produce a stepwise ladder around the mid; the simulated peak is at 0 ticks (real: -70), with correlation 0.8568 (Table 7).

| Series | JB p-value | LB $r_t$ p-value | LB $\lvert r_t \rvert$ p-value |
|---|---|---|---|
| Real holdout | 7.564e-07 | 0.2656 | 0.6726 |
| Simulator | 8.883e-16 | 0.3683 | 0.0927 |

Table 9: Distributional and dependence tests for daily holdout returns.

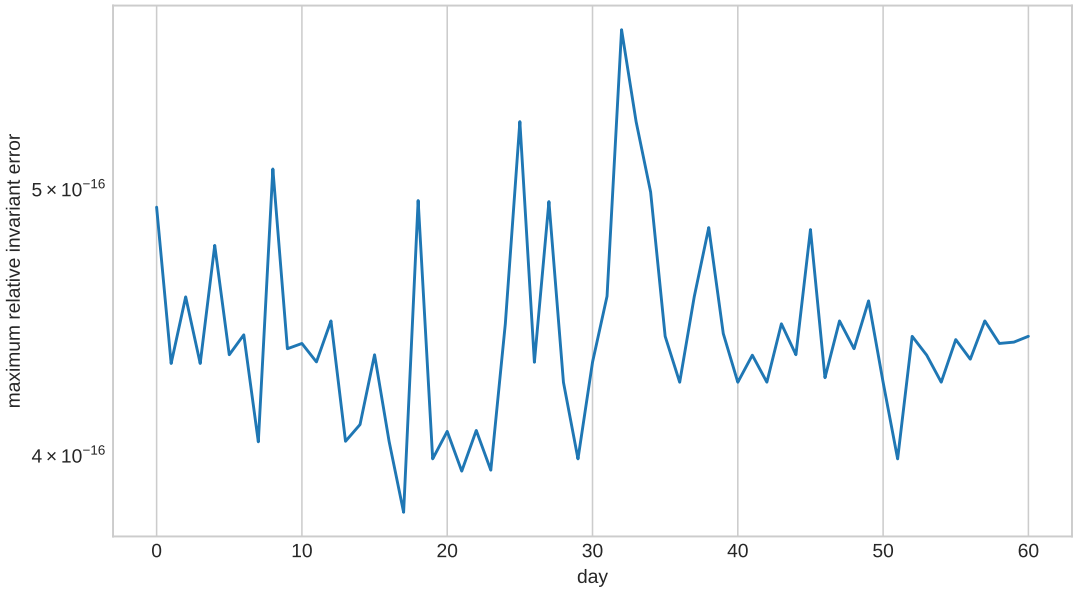



Figure 7: Maximum relative invariant error across open positions (semilog scale). Residuals remain at floating-point precision.

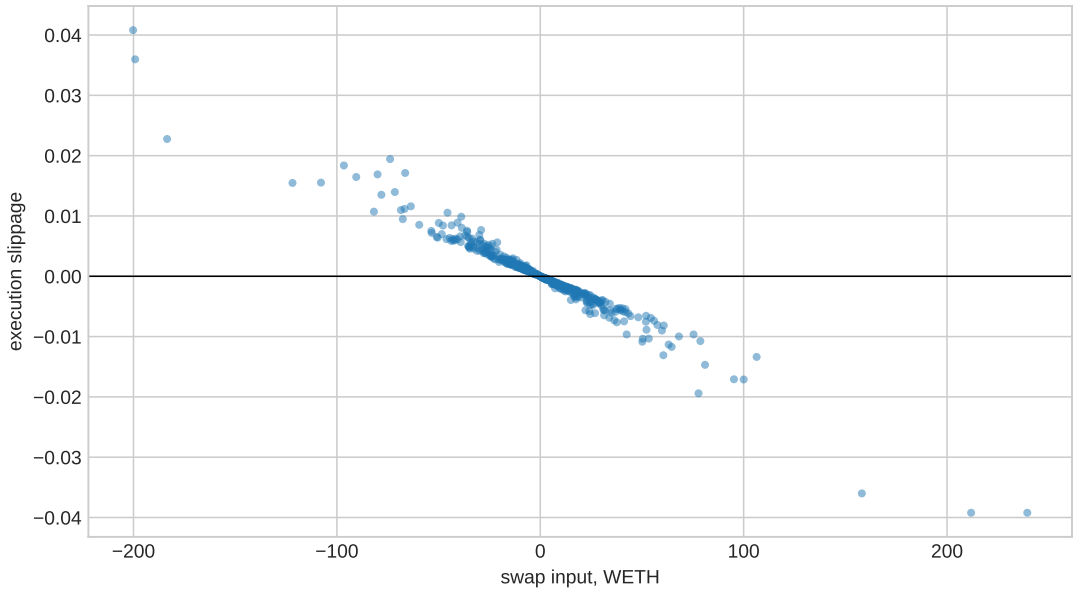



Figure 8: Swap flow and execution slippage. Slippage is small on average but increases when active liquidity is thin.

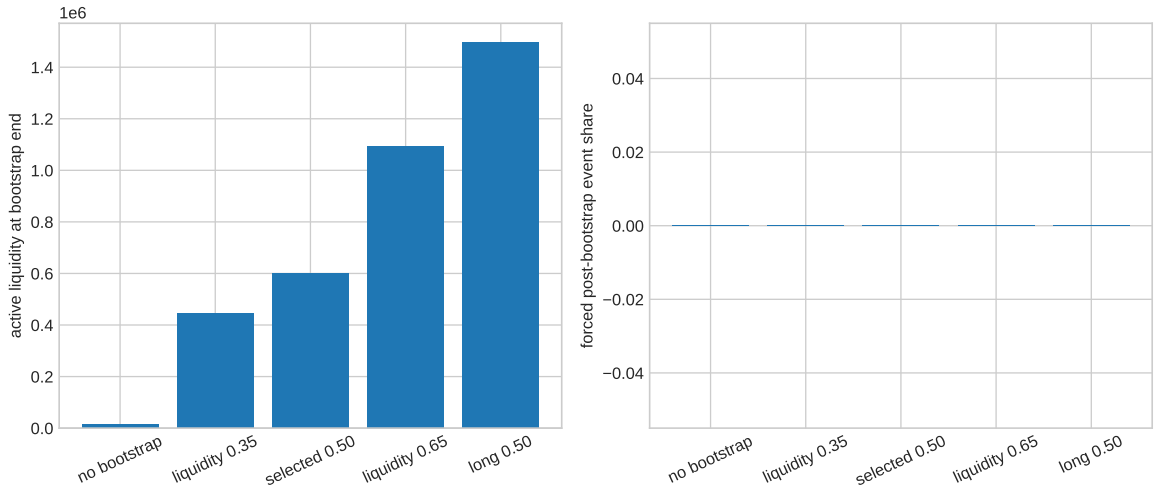



Figure 9: Sensitivity of the empty-pool initialisation to hypothetical high-mint bootstrap rules. The adopted event probabilities are the on-chain frequencies in Table 1.

# A Replication protocol and figure notes

This appendix specifies the computational protocol that produces every numerical claim. The event engine applies the position-level identities in Eqs. (1)-(9); the calibration layer estimates parameters from the training window; and the reporting layer regenerates all tables and figures from the frozen holdout paths. All floats below are pinned with the `[H]` specifier so that they cannot migrate into the reference list.

## A.1 Algorithmic skeleton

**Algorithm 1** One Monte Carlo path (`simulate`)

**Require:** $n$ events, seed, return parameters $(s, \nu, \lambda, \sigma_J)$, LP structure $(a, B, \tau_b, p_{\text{reb}}, q)$, mint-bundle mean $\bar{n}_{\text{LP}}$

Initialise empty position set; force first action = mint
**for** $t = 0, \ldots, n-1$ **do**
    Draw action from bootstrap or operating mixture (Table 1)
    **if** action = mint **then**
        Draw bundle size $N \sim \text{Poisson}(\bar{n}_{\text{LP}})$
        Mint $N$ nested/shoulder ranges; prune ranges farther than 1200 ticks and enforce the 350-position cap
    **else if** action = swap **then**
        Draw $r_t$ from Eq. (8); invert for $\Delta x_t$; update $\sqrt{P}$ via Eq. (7); allocate fees
        Refresh up to 2 inactive ranges toward the current tick
    **else if** action = burn **then**
        Sample position with weights (9); close and optionally re-mint a recentered micro-bundle
    **end if**
    **if** $t \bmod 8 = 0$ **then**
        Translate every core range to the current usable tick, preserving width, liquidity and fees; record old-range removal and new-range addition
    **end if**
    Record reserves, active/inactive $L$, slippage, invariant residual
**end for**
Aggregate every 8 events into one OHLCV bar

Replication uses the six-month event stream, the chronological split of 120 training bars and 61 holdout bars, and the parameters in Table 2. The reporting procedure then regenerates the tables and figures from 6 frozen paths. The compressed time resolution is 8 slots per daily bar and the mean mint bundle contains 12 legs.

## A.2 Figure-by-figure commentary

**Figure 1.** Holdout closes versus one Monte Carlo path after the train→test re-anchor. Diagnostic of scale only; success is distributional (Table 3).

**Figure 2.** Stage-1 screening heatmap and the refined training-only selection. Selected cell: $(s, \nu, \lambda, \sigma_J) = (0.028, 3.5, 0.500\%, 0.08)$.

**Figure 3.** Holdout versus simulated daily log-return histograms; complements Table 5.

**Figure 4.** Path-level $\widehat{\sigma}(r_t)$ across 6 seeds; dashed line is the real holdout value 0.0292.

**Figure 5.** Active versus inactive liquidity on the representative path after multi-LP minting and stranded-range pruning.

**Figure 6.** Normalised $L(\text{tick})$ reconstructed from the position-change ledger. Multi-LP nested mint bundles raise the number of nonzero steps from 217 simulated changes versus 668 empirical changes in the local window. The core sleeve is re-centred at every synthetic bar boundary, giving peak offset 0 ticks (real: -70) and correlation 0.8568.

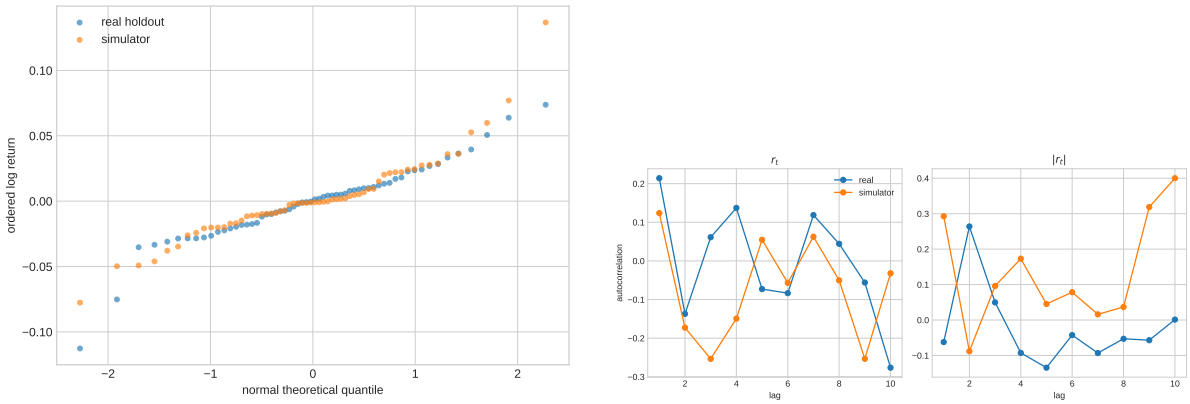
**Figure 7.** Maximum relative residual of Eq. (1); order $10^{-16}$.

**Figure 8.** Swap input versus execution slippage; dispersion widens when $L_{\text{act}}$ is thin.

**Figure 9.** Empty-pool bootstrap ablation; production probabilities remain those of Table 1.

## A.3 Supplementary diagnostic plots

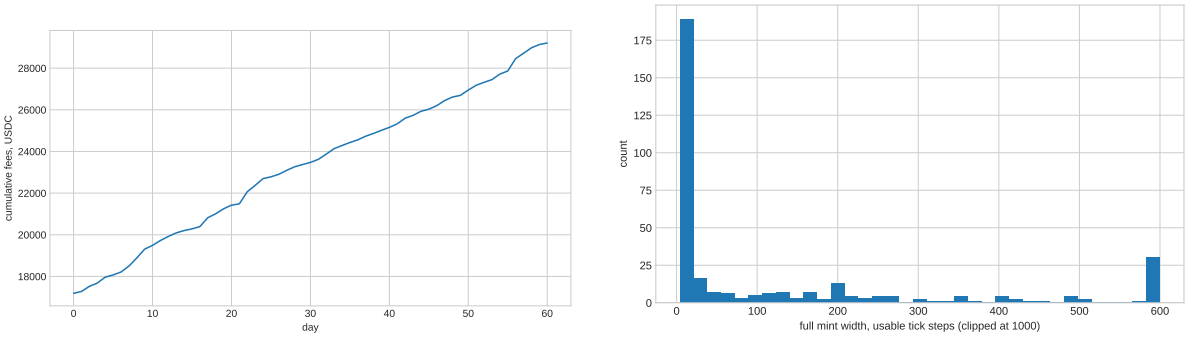
The following artefacts are retained for completeness. They are placed on dedicated pages so that they cannot interleave with the bibliography.




(a) Return QQ plot.

(b) Return / $|r_t|$ ACF.

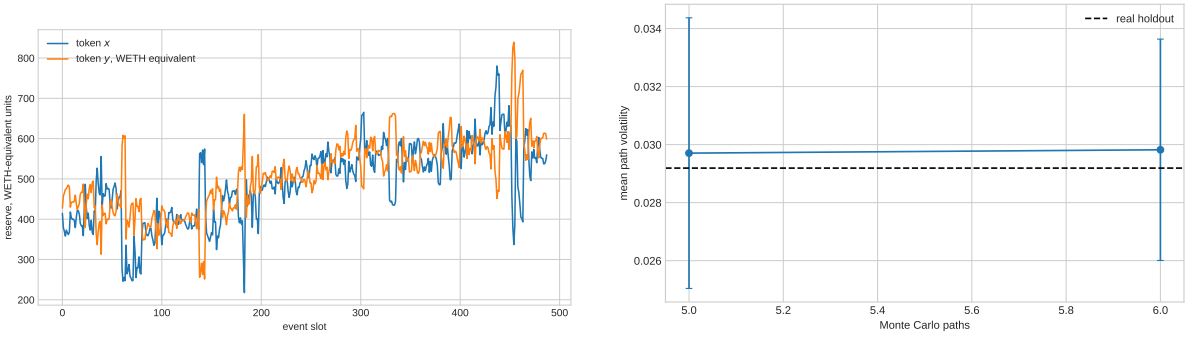
Figure 10: Supplementary return diagnostics. Panel (a) compares quantiles; panel (b) shows autocorrelations consistent with Table 9.




(a) Cumulative fees.

(b) Mint width law.

Figure 11: Fee accrual (a) and simulated mint-width distribution (b).




(a) Reserve decomposition.

(b) MC convergence.

Figure 12: Reserve decomposition from Eq. (4) (a) and Monte Carlo convergence of pooled volatility (b).

## A.4 Data products and numerical caveats

- OHLCV: daily open, high, low, close and dollar volume for the WETH/USDC 5 bps pool over the six-month observation window.

- Event stream: chronologically indexed mint, swap and burn records from the Ethereum mainnet Uniswap v3 subgraph.
- Position ledger: signed lower/upper tick changes used to reconstruct fixed-time range liquidity and active/inactive shares.
- Reported outputs: pooled Monte Carlo summaries, confidence intervals, distributional tests and the figures in the main text.

Known caveats from the code review:

1. Eight micro-events per bar undersample on-chain intensity; changing the compressed time resolution requires re-calibration.
2. Multi-LP mint bundles improve the stepwise geometry of Figure 6, with 217 simulated changes versus 668 empirical changes in the plotted window; further strategic LP heterogeneity remains open.
3. Pooled kurtosis need not lie inside the bootstrap interval of path-level kurtosis means.
4. Any alternative execution implementation must be reconciled against the position-level invariant before its outputs are pooled with these paths.

# B Additional protocol identities

Given deposit $\Delta x$ inside an active range,

$$\Delta L = \Delta x \frac{\sqrt{P}\sqrt{p_b}}{\sqrt{p_b} - \sqrt{P}}, \qquad \Delta y = \Delta L\big(\sqrt{P} - \sqrt{p_a}\big),$$

implemented directly by the simulator. Fee income on a swap is $\text{fee}_t = f|\Delta x_t|P_t$, allocated pro rata to positions with active ranges before the trade.

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